

Input data file for analysis and design using STAAD Pro is given here below:

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STAAD SPACE taxiSh
START JOB INFORMATION
ENGINEER DATE 26-Nov-19
END JOB INFORMATION
INPUT WIDTH 79
PAGE LENGTH 15000
SET NL 1000000
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 0 0 3; 3 4.5 0 0; 4 4.5 0 3; 5 9 0 0; 6 9 0 3; 7 13.5 0 0;
8 13.5 0 3; 9 0 3 -0.6; 10 0 3 0; 11 0 3 3; 12 1.5 3 -0.6; 13 1.5 3 3;
14 3 3 -0.6; 15 3 3 3; 16 4.5 3 -0.6; 17 4.5 3 0; 18 4.5 3 3; 19 6 3 -0.6;
20 6 3 3; 21 7.5 3 -0.6; 22 7.5 3 3; 23 9 3 -0.6; 24 9 3 0; 25 9 3 3;
26 10.5 3 -0.6; 27 10.5 3 3; 28 12 3 -0.6; 29 12 3 3; 30 13.5 3 -0.6;
31 13.5 3 0; 32 13.5 3 3; 33 0 5.2 2.5; 34 0 5.2 3; 35 0 5.2 3.5; 36 0 5.2 4;
37 0 5.2 4.5; 38 0 5.2 5; 39 0 5.2 5.5; 40 0 5.2 6.1; 41 1.5 5.2 2.5;
42 1.5 5.2 6.1; 43 3 5.2 2.5; 44 3 5.2 6.1; 45 4.5 5.2 2.5; 46 4.5 5.2 3;
47 4.5 5.2 3.5; 48 4.5 5.2 4; 49 4.5 5.2 4.5; 50 4.5 5.2 5; 51 4.5 5.2 5.5;
52 4.5 5.2 6.1; 53 6 5.2 2.5; 54 6 5.2 6.1; 55 7.5 5.2 2.5; 56 7.5 5.2 6.1;
57 9 5.2 2.5; 58 9 5.2 3; 59 9 5.2 3.5; 60 9 5.2 4; 61 9 5.2 4.5; 62 9 5.2 5;
63 9 5.2 5.5; 64 9 5.2 6.1; 65 10.5 5.2 2.5; 66 10.5 5.2 6.1; 67 12 5.2 2.5;
68 12 5.2 6.1; 69 13.5 5.2 2.5; 70 13.5 5.2 3; 71 13.5 5.2 3.5; 72 13.5 5.2 4;
73 13.5 5.2 4.5; 74 13.5 5.2 5; 75 13.5 5.2 5.5; 76 13.5 5.2 6.1;
77 1.5 3 0; 78 3 3 0; 79 6 3 0;
80 7.5 3 0; 81 10.5 3 0; 82 12 3 0;
83 1.5 5.2 3; 84 3 5.2 3; 85 6 5.2 3; 86 7.5 5.2 3; 87 10.5 5.2 3; 88 12 5.2 3;
MEMBER INCIDENCES
1 1 10; 2 2 11; 3 3 17; 4 4 18; 5 5 24; 6 6 25; 7 7 31; 8 8 32; 9 11 34;
10 18 46; 11 25 58; 12 32 70; 13 9 10; 14 10 11; 15 12 77; 16 14 78; 17 16 17;
18 17 18; 19 19 79; 20 21 80; 21 23 24; 22 24 25; 23 26 81; 24 28 82; 25 30 31;
26 31 32; 27 9 12; 28 12 14; 29 14 16; 30 16 19; 31 19 21; 32 21 23; 33 23 26;
34 26 28; 35 28 30; 36 11 13; 37 13 15; 38 15 18; 39 18 20; 40 20 22; 41 22 25;
42 25 27; 43 27 29; 44 29 32; 45 33 34; 46 34 35; 47 35 36; 48 36 37; 49 37 38;
50 38 39; 51 39 40; 52 41 83; 53 43 84; 54 45 46; 55 46 47; 56 47 48; 57 48 49;
58 49 50; 59 50 51; 60 51 52; 61 53 85; 62 55 86; 63 57 58; 64 58 59; 65 59 60;
66 60 61; 67 61 62; 68 62 63; 69 63 64; 70 65 87; 71 67 88; 72 69 70; 73 70 71;
74 71 72; 75 72 73; 76 73 74; 77 74 75; 78 75 76; 79 33 41; 80 41 43; 81 43 45;
82 45 53; 83 53 55; 84 55 57; 85 57 65; 86 65 67; 87 67 69; 88 40 42; 89 42 44;
90 44 52; 91 52 54; 92 54 56; 93 56 64; 94 64 66; 95 66 68; 96 68 76; 97 77 13;
98 78 15; 99 79 20; 100 80 22; 101 81 27; 102 82 29; 103 10 77; 104 77 78;
105 78 17; 106 17 79; 107 79 80; 108 80 24; 109 24 81; 110 81 82; 111 82 31;
112 83 42; 113 84 44; 114 85 54; 115 86 56; 116 87 66; 117 88 68; 118 34 83;
119 83 84; 120 84 46; 121 46 85; 122 85 86; 123 86 58; 124 58 87; 125 87 88;
126 88 70;
START GROUP DEFINITION
MEMBER
_LCOL 1 TO 8
_UCOL 9 TO 12
_MPBML 13 14 17 18 21 22 25 26
_SRAFTERL 15 16 19 20 23 24 97 TO 102
_EAVESL 27 TO 44
_MPBU 45 TO 51 54 TO 60 63 TO 69 72 TO 78
_SRAFTERU 52 53 61 62 70 71 112 TO 117
_TIELL 103 TO 111
_TIEU 118 TO 126
_EAVESU 79 TO 96
END GROUP DEFINITION
SUPPORTS
1 TO 8 PINNED
DEFINE MATERIAL START
ISOTROPIC S355
E 2.05e+008
POISSON 0.3
DENSITY 77
ALPHA 1.2e-005
DAMP 0.03
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TYPE STEEL
STRENGTH FY 35500 FU 510000 RY 1.5 RT 1.2
ISOTROPIC S275
E 2.05e+008
POISSON 0.3
DENSITY 77
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 27500 FU 430000 RY 1.5 RT 1.2
ISOTROPIC M40
E 3.33e+007
POISSON 0.2
DENSITY 25
ALPHA 1.2e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 40000
END DEFINE MATERIAL
START USER TABLE
TABLE 1
UNIT METER KN
*cut from RHS 250x150x12
CHANNEL
TC1
0.003912 0.15 0.012 0.1 0.012 1.34556e-005 3.81399e-006 1.87776e-007 -
0.0329939 0.0018 0.0016
TC2
0.004392 0.15 0.012 0.12 0.012 1.57466e-005 6.36529e-006 2.10816e-007 -
0.0414098 0.0018 .00192
TC3
0.004872 0.15 0.012 0.14 0.012 1.80377e-005 9.77728e-006 2.33856e-007 -
0.0501379 0.0018 0.00224
TC4
0.005352 0.15 0.012 0.16 0.012 2.03287e-005 1.41507e-005 2.56896e-007 -
0.0590942 0.0018 0.00256
TC5
0.005832 0.15 0.012 0.18 0.012 2.26197e-005 1.95849e-005 2.79936e-007 -
0.0682222 0.0018 0.00288
TC6
0.006312 0.15 0.012 0.2 0.012 2.49108e-005 2.61779e-005 3.02976e-007 -
0.0774829 0.0018 0.0032
END
*START GROUP DEFINITION
*MEMBER
*ELEMENT
*END GROUP DEFINITION
CONSTANTS
BETA 45 MEMB 79 TO 96
BETA 90 MEMB 1 TO 12 45 TO 51 54 TO 60 63 TO 69 72 TO 78
*MATERIAL M40 MEMB 1 TO 259
MATERIAL S355 all
MEMBER PROPERTY BRITISH
27 TO 44 79 TO 96 TABLE ST UA100X100X15
15 16 19 20 23 24 52 53 61 62 70 71 97 TO 102 112 TO 117 TABLE ST TUB80805.0
51 60 69 78 UPTABLE 1 TC1
50 59 68 77 UPTABLE 1 TC2
49 58 67 76 UPTABLE 1 TC3
48 57 66 75 UPTABLE 1 TC4
45 47 54 63 72 74 UPTABLE 1 TC5
13 14 17 18 21 22 25 26 46 55 56 64 65 73 UPTABLE 1 TC6
1 TO 12 TABLE ST TUB25015012.0
103 TO 111 118 TO 126 table st TUB80805.0
LOAD 1 LOADTYPE Dead TITLE DL
* DL on existing structure
SELFWEIGHT Y -1
floor load
yrange 2.99 3.01 fload -1

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yrange 5.19 5.21 fload -1
LOAD 2 LOADTYPE Live TITLE LLL
floor load
yrange 2.99 3.01 fload -.5
LOAD 3 LOADTYPE Live TITLE LLU
floor load
yrange 5.19 5.21 fload -.5
LOAD 4 LOADTYPE Wind TITLE WD
floor load
yrange 2.99 3.01 fload -.9
yrange 5.19 5.21 fload -.9
LOAD 5 LOADTYPE Wind TITLE WU
floor load
yrange 2.99 3.01 fload .9
yrange 5.19 5.21 fload .9
LOAD COMB 31 SLS1
1 1.0 2 1.0
LOAD COMB 32 SLS2
1 1.0 3 1.0
LOAD COMB 33 SLS3
1 1.0 2 1 3 1.0
LOAD COMB 34 SLS4
1 1.0 4 1.0
LOAD COMB 35 SLS5
1 1.0 5 1.0
LOAD COMB 51 ULS1
1 1.35 2 1.5
LOAD COMB 52 ULS2
1 1.35 3 1.5
LOAD COMB 53 ULS3
1 1.35 2 1.5 3 1.5
LOAD COMB 54 ULS4
1 1.35 4 1.5
LOAD COMB 55 ULS5
1 1.35 2 1.5 3 1.5 4 0.75
LOAD COMB 56 ULS6
1 1.0 5 1.5
PERFORM ANALYSIS PRINT STATICS CHECK
LOAD LIST 31 TO 34
PRINT JOINT DISPLACEMENTS
PRINT SUPPORT REACTION ALL
LOAD LIST 51 TO 55
PRINT MEMBER FORCES ALL
*print force envelope ns 10 list 3 54 105 160 211 248
PARAMETER 1
CODE EN 1993-1-1:2005
* NA 0:orig, 1: BS, 4:NF, 7:SS, 9:MS
NA 7 all
* sgr 3: S235, 1: S275, 2: S355, 3: S420, 4: S460
SGR 2 all
*LY 4.4 MEMB 9 TO 12
*LZ 4.4 MEMB 9 TO 12
*cmm 4 member 17 18
TRACK 2 all
fixed group
group mem 1 TO 8
group mem 9 TO 12
group mem 13 14 17 18 21 22 25 26
group mem 15 16 19 20 23 24 97 TO 102
group mem 27 TO 44
group mem 45 TO 51 54 TO 60 63 TO 69 72 TO 78
group mem 52 53 61 62 70 71 112 TO 117
group mem 103 TO 111
group mem 118 TO 126
group mem 79 TO 96
CHECK CODE all
*SELECT OPTIMIZED
PERFORM ANALYSIS PRINT STATICS CHECK

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CHECK CODE all
STEEL TAKE OFF
FINISH